

Graph Theory – Lecture Notes

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1 Introduction

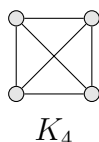
Definition 1.1. A (*simple*) *graph* G is a pair (V, E) where V is a non-empty set and $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$. We call the elements of V *vertices* of G and the elements of E *edges* of G . We also write $V(G)$ for V and $E(G)$ for E , respectively.

‘Simple’ means no loops and no double edges.

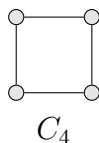
Remark 1.2. V is finite, unless otherwise stated.

Example 1.3. We define a few common graphs. Let $n, m \in \mathbb{N}$.

- The *complete graph* K_n has n vertices and each pair of vertices is connected.



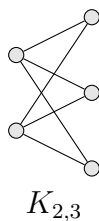
- The *cycle* C_n has n vertices, which are cyclically connected.



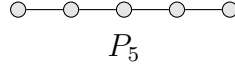
- Let A and B be disjoint sets with $|A| = n$ and $|B| = m$. The *complete bipartite graph* $K_{m,n}$ is defined by

$$V = A \dot{\cup} B \tag{1}$$

$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



- The *path* P_n has the vertices $1, \dots, n$ and edges $\{a, a+1\}$ for $a \in \{1, \dots, n-1\}$. (in books sometimes P_{n-1} has n vertices)



Definition 1.4. Let G be a graph, $u, v \in V(G)$

- If $\{u, v\} \in E(G)$, then u and v are **adjacent** and the edge $e = \{u, v\}$ is **incident** with both u and v .
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$ is the **neighbourhood** of v in G .
- $\deg_G(v) = |N_G(v)|$ is the **degree** of v .

Lemma 1.5 (Handshake Lemma). *Let G be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \quad (3)$$

(in particular the sum of the degree is even).

Proof. $\sum_{v \in V(G)} \deg_G(v) = \#\{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\} = 2|E(G)|$ □

Definition 1.6. Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists an isomorphism ϕ between them, i.e. a bijection $\phi : V \rightarrow V'$ such that $\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G')$

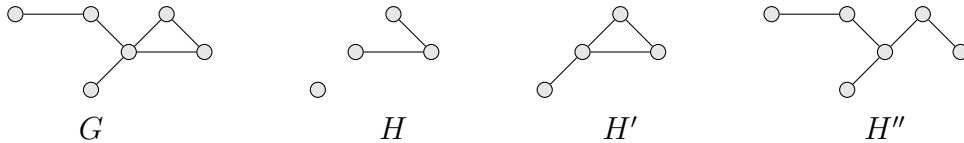
Definition 1.7. Let $G = (V, E)$ be a graph. A **subgraph** of G is a graph $H = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$ where $\binom{S}{K}$ is the set of all K -subsets of S .

H' is an **induced subgraph** of G if it is a subgraph and $E' = E \cap \binom{V'}{2}$

H'' is a **spanning subgraph** if it is a subgraph and $V' = V$

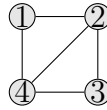
For $W \subseteq V$, write $G[W]$ for the subgraph induced by the vertices in W

Example 1.8.



Definition 1.9. Let v_i be vertices of a graph G

- A **walk** is a sequence of vertices v_i s.t. $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, \dots, k-1\}$
- The **length** of a walk is k (number of vertices in the walk)
- The walk is **closed** if $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which $v_1, \dots, v_k$ are pairwise distinct



$(1, 2, 3, 4, 2, 3)$ is a walk, but no trail

$(1, 2, 3, 4, 2)$ is a trail, no path

$(1, 2, 3, 4)$ is a path

Definition 1.10. Let G be a graph

- A graph G is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of G is an inclusion wise maximal connected subgraph

1.1 Hamiltonian Cycles

Definition 1.11. A **Hamiltonian cycle** in a graph G is a spanning subgraph that is a cycle (isomorphic to C_n for some $n \in \mathbb{N}$)

If G has a Hamiltonian cycle, G is called **Hamiltonian**

Theorem 1.12 (Dirac's Theorem). *Let G be a graph of order $n \geq 3$ where the **order** of G is given by $|V(G)|$ and the **size** by $|E(G)|$. If every Vertex of G has degree at least $\frac{n}{2}$ then G is Hamiltonian*

Theorem 1.13 (Ore's theorem). *Let G be a graph of order $n \geq 3$. If for every pair $u, v \in V, u \neq v, u$ and v non-adjacent, we have $\deg(u) + \deg(v) \geq n$ then G is Hamiltonian*

Remark 1.14.

Ore $\implies$ Dirac

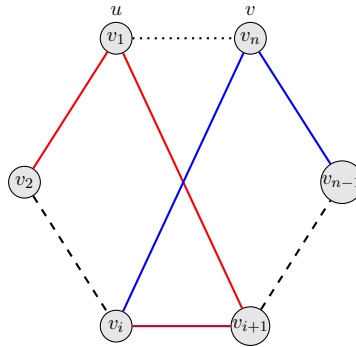
Proof.

The Graph G is connected: suppose G is not connected. Take an arbitrary component of G and call it H . The order of H is $n - k$ if k vertices are not in $V(H)$. Now any $v \in V(H)$ has $\deg(v) \leq n - k - 1$ with equality if v is adjacent to every vertex in $V(H)$. Now take $u \notin V(H)$ so $\deg(u) \leq k - 1$. It follows that $u, v \in V(G), u \neq v$ and u, v are not adjacent so $\deg(u) + \deg(v) \geq n$ but $\deg(u) + \deg(v) \leq k - 1 + n - k - 1 = n - 2 < n$ $\nmid$

- G has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge $\implies G$ complete, hence Hamiltonian
- Assume G non complete $u, v \in V, u \neq v$, be non-adjacent. By induction the graph G' obtained from G by adding $e = \{u, v\}$ as an edge is Hamiltonian. Let C be a Hamiltonian cycle of G' . If C does not use $e \implies C$ is Hamiltonian cycle in G . Assume C uses e . Removing e from C yields a path from u to v

For $i \in \{2, \dots, n - 1\}$

- If v_i is a neighbour of u , color $\{v_{i-1}, v_i\}$ red
- If v_i is a neighbour of v , color $\{v_i, v_{i+1}\}$ blue
- Since $\deg(u) + \deg(v) \geq n$ and the path $v_1, \dots, v_n$ has $n - 1$ edges, there exists an edge $\{v_i, v_{i+1}\}$ which is both red and blue. Then $v_1, \dots, v_i, v_n, v_{n-1}, \dots, v_{i+1}$ is a Hamiltonian cycle in G .



□

1.2 Eulerian Trail

Definition 1.15. An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

Theorem 1.16 (Euler's theorem). *Let G be a connected graph with at least 2 vertices. Then G has a closed Eulerian trail iff every vertex in G has even degree*

Proof.

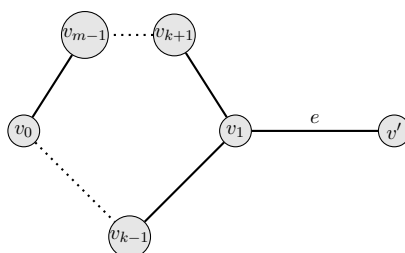
" $\implies$ "

Let T be a closed Eulerian trail and fix an orientation on T . Then for every $v \in V(G)$, the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of T , so $\deg(v)$ is even.

" $\impliedby$ "

Let $T = (v_0, e_1, v_1, \dots, e_m, v_m)$ be a longest trail. We will prove (i) $v_0 = v_m$ and (ii) $\{e_i : i \in \{1, \dots, m\}\} = E(G)$. (i)+(ii) $\iff$ T is closed Eulerian trail.

- (i) If $v_0 \neq v_m$, then v_0 is incident to an odd number of edges in T . As $\deg(v_0)$ is even, there is $e \in E(G)$ incident to v_0 and not on T . We could thus append e on T $\nmid$ because T is longest trail.
- (ii) Claim: $V(T) = V$. Assume otherwise there exists a vertex $v' \notin V(T)$ and $e = \{v_k, v'\} \in E(G)$ for some $k \in \{0, \dots, m\}$ (by connectedness). Then $(v', e, v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, \dots, v_{k-1}, e_k)$ is a longer trail hence $V(T) = V$



Claim: $E(T) = E$. Assume otherwise. Consider $e \in E \setminus E(T)$ and write $e = \{v_k, v_l\}$. Then $(v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, v_1, \dots, e_k, v_k, e, v_l)$ is a longer trail $\nmid$

□

Theorem 1.17. *Let G be a connected graph with at least two vertices. Then G has an Eulerian trail iff there are precisely 0 or two vertices of odd degree*

Proof.

By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

Suppose now that G has exactly two vertices x and y of odd degree. We consider the following graph G' : Set $V(G') = V(G) \cup \{z\}$, where $z \notin V(G)$. Furthermore set $E(G') = E(G) \cup \{\{x, z\}, \{y, z\}\}$. Clearly, every vertex has even degree in G' , so there exists a closed Eulerian trail. Since z is only adjacent to x and y , these are the neighbours of z on the trail. Deleting the edges $\{x, z\}$ and $\{y, z\}$ from the trail yields an Eulerian trail from x to y in G . Conversely, if there is a non-closed Eulerian trail with endpoints x and y in G , then x and y are non-adjacent. Adding the edge $\{x, y\}$ creates a graph with a closed Eulerian trail, so all vertices have even degree by Euler's theorem. Thus x and y are the only vertices in G with odd degree. $\square$

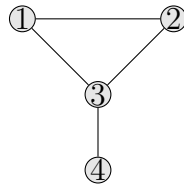
1.3 Datastructures

Assume that $G = (V, E)$ is a graph with $v = \{1, \dots, n\}$

Definition 1.18. The **adjacency matrix** of G is the matrix $A = (a_{ij})$ with
$$a_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & \text{else} \end{cases}$$

The **adjacency list** of G stores a label for each vertex, and a list of (pointers to) its neighbours.

Example 1.19.



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- v_1 : v_2, v_4

- v_2 : v_1, v_3, v_4
- v_3 : v_2
- v_4 : v_1, v_2

Remark 1.20.

Memory requirement:

Adjacency matrix: $\theta(n^2)$

Adjacency list: $\theta(n + m)$, where m is the number of edges

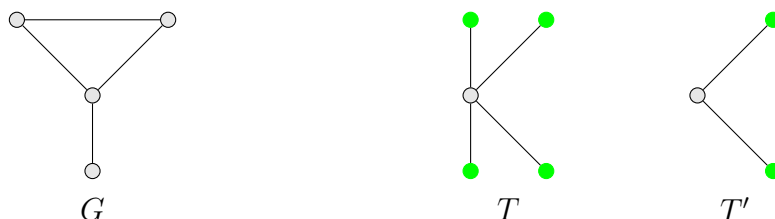
2 Trees

Definition 2.1. A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to C_n for some $n \geq 3$).

A **forest** is a graph that does not contain a cycle.

A **leaf** is a vertex of a tree with degree 1.

Example 2.2.



- G is not a tree nor a forest
- T, T' are trees and forests
- $T \cup T'$ is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

Lemma 2.3. *Every tree T with $n \geq 2$ vertices has at least one leaf.*

Proof. We iteratively construct a path $v_1, \dots, v_n$ as follows:

Start with any edge $\{v_1, v_2\} \in E(T)$. Assume that we have constructed $v_1, \dots, v_k$. To construct v_{k+1} , we may assume that v_k is not a leaf in T (or we are done). Otherwise v_k has a neighbour $v_{k+1} \neq v_{k-1}$. $v_{k+1} \neq v_i$ because T is a tree and otherwise we would get a cycle $(v_i, \dots, v_k, v_{k+1} = v_i)$. Add v_{k+1} to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf. $\square$

Theorem 2.4. *Let G be a graph with n vertices. The following are equal:*

- G is a tree
- G is connected with $n - 1$ edges
- G is connected and removing an edge disconnects it
- G has exactly one path between each pair of distinct vertices

(v) G has no cycle, but adding an edge creates a cycle

Proof. "1. $\implies$ 2."

Induction on n :

$n = 1$ is trivial

Let $n \geq 2$. By Lemma (2.2), G has a leaf v . Removing v and the unique edge incident to v yields a tree G' . By induction G' has $n - 2$ edges so G has $n - 1$ edges.

"2. $\implies$ 3."

Suppose there is an edge $e \in E(G)$ and the resulting graph by removing e is still connected. Keep removing as long as the resulting graph is still connected. Let G' be the resulting graph. Observe that G' is a spanning subgraph of G . G' does not contain a cycle as otherwise we could remove one edge from the cycle. By "1. $\implies$ 2.", applied to G' , we know that G' has $n - 1$ edges $\nless$ because G had $n - 1$ edges and we removed at least one.

"3. $\implies$ 4."

Suppose that there is a pair with $u, v \in V(G)$ s.t. there are two distinct paths u_i, v_i between them. Let i_0 be the maximum index s.t. $u_i = v_i : \forall i \in \{1, \dots, i_0\}$. The edge $\{v_{i_0}, v_{i_0+1}\}$ does not lie on the path u_i . The graph obtained by removing $\{v_{i_0}, v_{i_0+1}\}$ is connected as for any walk using this edge we can use u_i to go to u to reach v_{i_0} or v to reach v_{i_0+1} .

"4. $\implies$ 5."

If G had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).

Adding an edge creates a cycle: Let $u, v \in V(G)$ non adjacent. There is one path between the two. Now adding an edge between u, v creates a second path $\nless$

"5. $\implies$ 1."

We only need to show that G is connected. Suppose u, v are not connected by a walk. Creating an edge between u, v does not result in a cycle because for being isomorphic to C_n one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting u, v is the newly added edge $\nless$ $\square$

Corollary 2.5. *Every connected graph G has a spanning tree.*

Proof.

Consider graphs on n vertices, now do induction on the number of edges m . As G

is connected, $m \geq n - 1$. If $m = n - 1 \implies G$ is a tree. If $m > n - 1 \implies G$ is not a tree. By (2.3) we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of G . $\square$

Theorem 2.6 (Matrix-tree theorem). *Let G be a graph with vertices $\{1, \dots, n\}$, edges $\{e_1, \dots, e_m\}$.*

Let $Q = (q_{ij})$ with $q_{ij} = \begin{cases} -1 & \{i, j\} \in E \\ 0 & i \neq j : \{i, j\} \notin E \\ \deg(i) & i = j \end{cases}$ Let Q_{ij} be the matrix obtained from Q by removing the i -th row and the j -th column. Then the number of labelled spanning trees of G is $|\det(Q_{ij})|$, for any choice of i, j .

Corollary 2.7 (Cayley's formula). *The number of labelled spanning trees of K_n is n^{n-2} if $n \geq 2$.*

Proof. Define Q as in (2.5).

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \cdots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \cdots & 1 & n-1 \end{pmatrix} \rightarrow \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -n & n & 0 & \cdots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \cdots & 0 & n \end{pmatrix} \rightarrow \hat{Q}_{11} =$$

$$\begin{pmatrix} 1 & \star & \cdots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \cdots & 0 & n \end{pmatrix}$$

$$\implies |\det(Q_{11})| = |\det(\hat{Q}_{11})| = n^{n-1-1} = n^{n-2} \quad \square$$

3 Planar Graphs

Definition 3.1. A **planar drawing** of a graph G consists of injective continuous functions

$$\phi_v : V(G) \rightarrow \mathbb{R}^2 \qquad \phi_e : [0, 1] \rightarrow \mathbb{R}^2$$

s.t.:

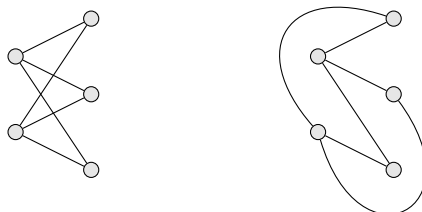
- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$ for all $\{u, v\} \in E(G)$
- The sets $\phi_e((0, 1))$ for each $e \in E(G)$ are pairwise disjoint and disjoint from $\text{Im}(\phi_v)$ (i.e. no crossings)

A graph G is called **planar** if it has a planar drawing.

A **plane graph** is a planar graph, together with a fixed planar drawing.

Example 3.2.

- Trees
- K_4
- $K_{2,3}$



Definition 3.3. A **face** of a plane graph G is a maximal connected part of $\mathbb{R}^2 \setminus (\text{Im}(\phi_v(u)) \cup \text{Im}(\phi_e))$. Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of G incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

Theorem 3.4 (Euler's formula). *Let $G = (V, E)$ be a connected planar graph. Then the number of faces $|F|$ of G is $|F| = |E| - |V| + 2$.*

Proof. Induction on $|E|$.

Since G is connected, we have $|E| \geq |V| - 1$ so we start the induction at $|E| = |V| - 1 \implies G$ is a tree. Since a tree does not contain a cycle, we have $|F| = |E| - |V| + 2$. Now let $|E| > |V| - 1$. Then G is not a tree, so G contains a cycle C . Choose an edge e on C and remove it to obtain a plane graph G' . e separates two faces in G , which fuse to a single face in G' . By induction, the number of faces $|F'|$ in G' is $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$. $\square$

Lemma 3.5. *For every plane graph $G = (V, E)$ with $|V| \geq 3$, we have $|E| \leq 3|V| - 6$.*

Proof. Wlog G is connected. Consider a planar drawing of G and let F be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula (3.4), we obtain $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$. $\square$

Remark 3.6. This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

Example 3.7.

- K_5 is not planar (10 edges) due to (3.4)
- $K_{3,3}$ is not planar (without proof)

Definition 3.8. A **subdivision** of a graph G is a graph obtained from G by repeatedly subdividing some of its edges.

Theorem 3.9 (Kuratowski's theorem). *A graph G is planar iff G contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.*

Remark 3.10. A subdivision of a graph is planar iff the graph is planar

Definition 3.11. Let G be a graph. The graph obtained from G by contracting $e = \{u, v\} \in E(G)$ denoted G/e , is the graph obtained from G by deleting u, v , inserting a new vertex w and making w adjacent to all vertices in $N_G(u) \cup N_G(v)$. A graph G' is a **minor** of G if G' can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

Remark 3.12. If G' is a minor of G then G' planar $\iff G$ is planar.

Theorem 3.13 (Wagner's theorem). *A graph G is planar iff $K_{3,3}$ and K_5 are not a minor of G .*

4 Flows, Cuts, Connectivity

Definition 4.1. A **directed graph** is a pair (V, E) with $E \subseteq (V \times V) \setminus \{(x, x) : x \in V\}$. Write $\delta_G^+(v)$ for edges starting in v and $\delta_G^-(v)$ for edges ending in v .

Definition 4.2. A **flow network** is a tuple (G, s, t, c) where G is a **directed** graph, $s, t \in V(G)$, $s \neq t$ and $c : E \rightarrow \mathbb{R}_0^+$.

- s is called a **source**
- t is called a **target**
- $\forall e \in E(G) : c(e)$ is the **capacity of e**

A **flow** in a flow network is a function $f : E \rightarrow \mathbb{R}_0^+$ s.t.

- $f(e) \leq c(e)$ for all $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$ (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$ (i.e. nothing flows into a source or out of a target)

The **value** of a flow f is $\text{val}(f) = \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$

Remark 4.3. For $v \in V$, write $f(\delta^+(v)) = \sum_{e \in \delta^+(v)} f(e)$; $f(\delta^-(v)) = \sum_{e \in \delta^-(v)} f(e)$

Definition 4.4. A **cut** in a graph/directed graph G is a partition of $V(G)$ into two non-empty parts A and B (i.e. $V(G) = A \dot{\cup} B$ and $A, B \neq \emptyset$). For $s, t \in V(G)$, $s \neq t \implies (A, B)$ is a (s, t) -cut if $s \in A, t \in B$.

If (S, T) is a (s, t) -cut, write $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$ and $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$.

The **capacity** of the cut is $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$.

Theorem 4.5 (max flow-min cut theorem). *Let (G, s, t, c) be a flow network. Then $\max\{\text{val}(f) : f \text{ flow}\} = \min\{c(S, T) : (S, T) \text{ is a } (s, t)\text{-cut}\}$.*

Lemma 4.6. *Let f be a flow, (S, T) be a (s, t) -cut. Then $\text{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$.*

Proof.

Claim: $\forall A \subset V : f(\delta^+(A)) - f(\delta^-(A)) = \sum_{v \in A} f(\delta^+(v)) - f(\delta^-(v)) = (*)$

The sum $f(\delta^+(A)) - f(\delta^-(A))$ does not account for edges $e \in E(G)$ with start and end in A . On the right side, $f(e)$ is counted twice, once with positive and once with negative sign, so the sum vanishes.

Then $\text{val}(f) = f(\delta^-(t)) \stackrel{(1)}{=} \sum_{v \in T} f(\delta^-(v)) - f(\delta^+(v)) \stackrel{(*)}{=} f(\delta^-(T)) - f(\delta^+(T)) = f(\delta^+(S)) - f(\delta^-(S))$

(1) holds because $f(\delta^+(t)) = 0$ (t is a sink) and $\forall v \notin \{s, t\} : f(\delta^-(v)) - f(\delta^+(v)) = 0$ (by definition). $\square$

Corollary 4.7. *Let f be a flow, (S, T) an (s, t) -cut.*

$\text{val}(f) \leq c(\delta^+(S))$

Thus $\max\{\text{val}(f) : f \text{ flow}\} \leq \min\{c(S, T) : (S, T) \text{ } (s, t)\text{-cut}\}$.

4.5.

$\max \text{ flow} \leq \min \text{ cut}$ due to Corollary (4.7)

Let f maximum flow (i.e. a flow for which $\text{val}(f)$ is maximal). Let $c' : E(G) \rightarrow \mathbb{R}_0^+$ where $c'(e) = c(e) - f(e)$. Let $S \subset V(G)$ be defined as the set of reachable vertices¹ according to the following rules:

- $s \in S$
- $x \in V(G)$ is in S (i.e. reachable) if there is a walk (in the underlying directed graph) from s to x traversing $e_1, \dots, e_k$ s.t.
 - (i) $c'(e_i) > 0$ ($\iff f(e_i) < c(e_i)$) if e_i is traversed towards x .
 - (ii) $c'(e_i) < c(e_i)$ ($\iff f(e_i) > 0$) if e_i is traversed towards s .

If $S = V(G)$, we can improve f as follows:

Let $e_1, \dots, e_k$ be a walk from s that makes t reachable, and e_i is traversed in orientation $\sigma_i \in \{\pm 1\}$, then $\tilde{f}(e_i) = f(e_i) + \sigma_i \varepsilon$, for $i \in \{1, \dots, k\}$ and $\tilde{f}(e_i) = f(e_i)$ otherwise. This defines a flow $\tilde{f}$ with $\text{val}(\tilde{f}) > \text{val}(f)$ for $\varepsilon > 0$ sufficiently small $\nmid$ Thus $t \notin S$. Set $T = V \setminus S \neq \emptyset$

- $\forall e \in \delta^+(S) \quad c'(e) = 0$, i.e. $f(e) = c(e)$
 $e = \{x, y\}$ with $x \in S, y \in T$. If $c'(e) \neq 0 \implies e$ satisfies (i) $\implies y$ reachable $\nmid$

¹These are the vertices where the flow "is not maximal yet" but has a walk to s .

- $\forall e \in \delta^-(S) \quad c'(e) = c(e) \iff f(e) = 0$
- $e = \{y, x\}$ with $x \in S, y \in T$. If $c'(e) \neq c(e) \implies e$ satisfies (ii) $\implies y$ reachable $\nmid$
- $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = \sum_{e \in \delta^+(S)} f(e) - \sum_{e \in \delta^-(S)} f(e) = f(\delta^+(S)) - f(\delta^-(S)) \stackrel{(4.6)}{=} \text{val}(f) \implies \text{val}(f) \geq c(S, T) \geq \min(s, t)\text{-cut}$

□

4.1 Connectivity

Definition 4.8. A graph G is **k -vertex connected** ($k \in \mathbb{N}$) if it has at least $k + 1$ vertices and whenever we remove up to $k - 1$ vertices, the resulting graph is connected.

A graph G is **k -edge connected** if whenever we remove up to $k - 1$ edges, the resulting graph is connected.

The **vertex connectivity (edge connectivity)** is the largest $k \in \mathbb{N}$ s.t. G is k -vertex/edge connected.

Remark 4.9. G graph, k vertex connectivity, λ edge connectivity, δ minimal degree of G , then $k \stackrel{(i)}{\leq} \lambda \stackrel{(ii)}{\leq} \delta$

Proof. (i) If we have a set of λ edges whose removal makes the res. graph disconnected, then deleting one endpoint for each edge also disconnects the graph.

(ii) Delete all edges incident to a vertex of minimal degree, this disconnects the graph.

□

Lemma 4.10. G directed graph, $s, t \in V, s \neq t$. Then the maximal number of edge-disjoint (s, t) -paths (path which do not have any common edges) is equal to the size of a minimal (S, T) -cut.

Proof. "Max flow-min cut theorem" (4.5) with $\forall e \in E \quad c(e) = 1$.

□

Theorem 4.11 (Menger's theorem for edge-connectivity). Let G be a graph, $k \in \mathbb{N}$. G is k -edge-connected iff there are k edge-disjoint paths between any pair of vertices in G .

Proof.

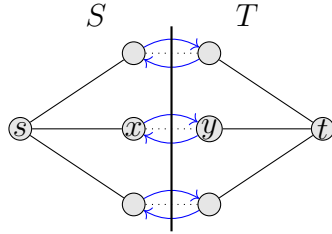
" $\Leftarrow$ " Removing $k - 1$ edges will keep one edge intact, so the resulting graph is connected, hence G is k -edge connected.

" $\Rightarrow$ " Let $s, t \in V, s \neq t$. Let G' be the graph obtained from G by replacing each edge by a double-edge (see graph(1)).

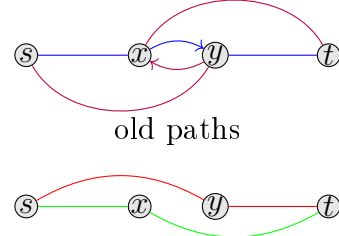
Claim: Any (s, t) -cut in G' has at least size k .

If (S, T) is an (s, t) -cut of size $\leq k - 1$, then G has at most $k - 1$ edges between S and T . Removing them disconnects the graph $\nmid G$ is k -edge connected. By Lemma (4.10), there are k -edge disjoint paths $p_1, \dots, p_k$ from s to t in G' .

Suppose $x, y \in V$ s.t. p_i uses $\{x, y\}$ and p_j uses $\{y, x\}$. Replace p_i and p_j by the path following p_i to x and continue via p_j and the path following p_j to y and continue via p_i . Repeat for each such edge. In each step, the sum of the lengths of the paths decreases so the process has to terminate. Then the underlying k -undirected paths in G are edge disjoint (see graph(2)).



graph(1)



new paths

graph(2)

□

Theorem 4.12 (Menger's theorem for vertex-connectivity). *Let G be a graph with at least $k + 1$ vertices ($k \in \mathbb{N}$). Then G is k -vertex connected iff there are at least k internally disjoint paths (start and end are the only common vertices) between any pair of distinct vertices of G .*

Proof. " $\Leftarrow$ " As before (Proof of (4.11)).

" $\Rightarrow$ " Let $a, b \in V$ be distinct and $v_1, \dots, v_n$ be the remaining vertices of G . Consider the following directed Graph G' (see graph(1)):

- $V(G') = \{s, t, v'_1, \dots, v'_n, v''_1, \dots, v''_n\}$
- $E(G') = \{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\} \cup \{\{v''_i, v'_j\} : \{v_i, v_j\} \in E(G)\} \cup \{\{v''_j, v'_i\} : \{v_i, v_j\} \in E(G)\} \cup \{\{s, v'_i\} : \{a, v_i\} \in E(G)\} \cup \{\{v''_i, t\} : \{b, v_i\} \in E(G)\} \cup \{\{s, t\} \text{ if } \{a, b\} \in E(G)\}$ where we call elements of $\{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\}$ red edges.

Let k_0 be the minimum size of an (s, t) -cut in G' . If $k_0 \geq k$, G' has k -edge disjoint paths from s to t . Since no red edge is contained in two of those, this corresponds to k internally disjoint paths from a to b in G : each path is of the form $(s, v'_1, v''_1, \dots, v'_m, v''_m, t)$ and corresponds to $(a, v_i, \dots, v_m, b)$ in G .

Now assume $k_0 < k$ and consider an (s, t) -cut (S, T) of G' of size k_0

Claim: We may assume that every edge in $\delta^+(S) \setminus \{s, t\}$ is red:

If an edge entering v'_i is in $\delta^+(S)$, move v'_i from T to S without changing the size of the cut. Similarly if an edge leaving v''_i is in $\delta^+(S)$, move v''_i to T . If $\{s, t\} \notin E(G)$, then removing the k_0 vertices of G that correspond to red edges in $\delta^+(S)$ disconnects G (no path from s to t in G' after removal $\implies$ none from a to b in G) $\nmid$.

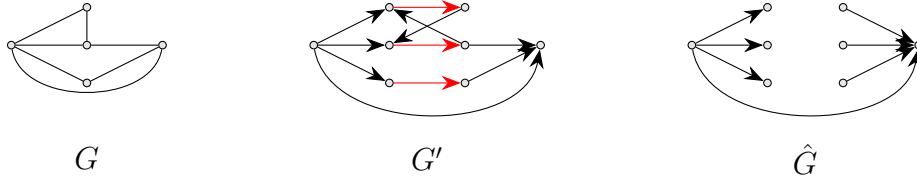
If $\{s, t\} \in E(G)$, proceed as follows:

Let A be the set of vertices in G corresponding to red edges in $\delta^+(S)$ in G' ($|A| = k_0 - 1$). Let $\hat{G}$ be the graph obtained by removing A .

If $\hat{G}$ not connected $\nmid G$ is k -vertex connected.

Thus $\hat{G}$ connected. Since G has $\geq k + 1$ vertices either a or b has an additional neighbour.

Wlog a does $\implies \hat{G} - a$ is not connected $\nmid G$ is k -vertex connected (see graph(2)).



□

Definition 4.13. A graph is **bipartite** with parts A, B if $V(G) = A \cup B$ and every edge in G has one end in A and one in B .

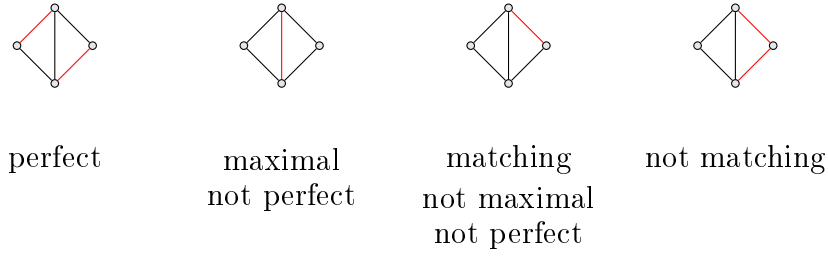
Example 4.14. • $K_{a,b}$

• $Q_d, d \geq 1$

Definition 4.15. A **matching** in a graph G is a set $M \subset E(G)$ s.t. no two edges in M are incident.

A matching M is **perfect** if every vertex in G is incident to an edge in M .

Example 4.16.



Theorem 4.17 (Hall's marriage theorem). *Let $G = (V, E)$ be a bipartite graph with part A, B . Then G has a perfect matching iff $\forall P \subset A, |N_G(P)| \geq |P|$ and $\forall P \subset B, |N_G(P)| \geq |P|$.*

Proof. " $\implies$ " Assume (Wlog) $\exists P \subset A$ s.t. $|N_G(P)| < |P|$. Let M be a perfect matching. Then $\forall v \in P$ the other endpoint of the matching edge incident to v is in $N_G(v) \subset N_G(P)$, and these vertices are distinct $\implies |N_G(P)| \geq |P|$. $\nmid$

" $\impliedby$ " Note that the condition implies $|A| = |B|$, since $|B| \geq |N_G(A)| \geq |A|$ and vice versa. Define an auxiliary graph $G' = (V', E')$ with:

- $V' = V \cup \{s, t\}$
- $E' = \{\{s, a\} : a \in A\} \cup \{\{b, t\} : b \in B\} \cup \{\{a, b\} : a \in A, b \in B\}$
- $\forall e \in E : c(e) = 1$

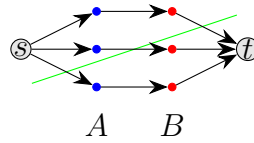
Every matching M in G defines an integral (s, t) -flow ($f(e) \in \{0, 1\}$) in G' with $\text{val}(f) = |M|$ and vice versa.

Let (S, T) be an (s, t) -cut. Let $A_S = A \cap S, A_T = A \cap T, B_S = B \cap S, B_T = B \cap T$. It follows $|\delta^+(S)| \geq |A_T| + |B_S| + |N_{A_S}(B_T)| \geq |N_{A_T}(B_T)| + |B_S| + |N_{A_S}(B_T)| =$

$$|N_A(B_T)| + |B_S| \stackrel{\text{condition}}{\geq} |B_T| + |B_S| = |B| = |A|.$$

Thus by maxflow-mincut, $\max_{f \text{ flow}} \text{val}(f) = \max_{f \text{ integral flow}} \text{val}(f) \geq |B|$.

Therefore there exists a matching in G of size $\geq |B|$, i.e. a perfect matching.



□

5 Coloring

Definition 5.1. Let $k \in \mathbb{N}$. A (proper) **k -vertex-coloring** (k -coloring) of a graph G is a function $c : V(G) \rightarrow \{1, 2, \dots, k\}$ such that for every edge $\{u, v\} \in E(G)$, we have $c(u) \neq c(v)$. The **chromatic number** $\chi(G)$ of a graph G is the smallest integer k such that G has a proper k -coloring. We call G **colorable** if it has a k -vertex-coloring.

Write $\Delta(G)$ for the maximum degree of a vertex in G .

Lemma 5.2. *Let G be a graph. Then $\chi(G) \leq \Delta(G) + 1$.*

Proof. Order the vertices of G as $v_1, v_2, \dots, v_n$. We will color the vertices in this order. When we come to color v_i , it has at most $\Delta(G)$ neighbors, so there are at most $\Delta(G)$ colors that are forbidden for v_i . Since we have $\Delta(G) + 1$ colors available, there is always at least one color that we can use to color v_i . $\square$

Remark 5.3. The bound $\chi(G) \leq \Delta(G) + 1$ is tight with equality for exactly the complete graphs K_n and the odd cycles C_{2k+1} .

Lemma 5.4. *Let G be a (connected) graph. If there is $v \in V(G)$ such that $\deg_G(v) < \chi(G)$, then G has a coloring with $\Delta(G)$ colors.*

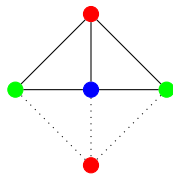
Proof. Proof by Induction on $|V(G)|$.

Base case: If $|V(G)| = 1$, then $\chi(G) = 1$ and $\Delta(G) = 0$, so the statement holds.

Induction step: Assume the statement holds for all graphs with fewer than n vertices, and let G be a graph with n vertices. Let $v \in V(G)$ such that $\deg_G(v) < \chi(G)$. Consider the graph $G' = G - v$. Since G' has fewer vertices than G , by the induction hypothesis, we have $\chi(G') \leq \Delta(G')$. Now let's look at two cases:

(1) $\exists w \in V(G)$ such that $\deg_G(w) = \Delta(G)$ and w is a neighbour of v in G and there is no $w' \in V(G')$ with $\deg_{G'}(w') = \Delta(G)$ not a neighbour of v . Then it follows that $\Delta(G) = \Delta(G') + 1 = \chi(G') + 1$ and $\chi(G) = \chi(G') + 1$. Therefore $\Delta(G) = \chi(G)$.

(2) There is no $w \in V(G)$ with $\deg_G(w) = \Delta(G)$ s.t. w is a neighbour of v in G . Then $\Delta(G) = \Delta(G')$ and $\chi(G) = \chi(G')$. Therefore $\Delta(G) = \chi(G)$.



$\square$

Theorem 5.5 (Brooks' Theorem). *Let G be a connected graph. If G is neither an odd cycle nor a complete graph, then $\chi(G) \leq \Delta(G)$.*

Proof. $\Delta(G) = 0$: Then G is a single vertex and therefore complete.

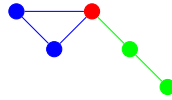
$\Delta(G) = 1$: Then G is a path and therefore complete.

$\Delta(G) = 2$: Then G is a cycle. If G is an odd cycle, then $\chi(G) = 3 = \Delta(G) + 1$. If G is an even cycle, then $\chi(G) = 2 = \Delta(G)$.

Let $\Delta(G) \geq 3$

By Lemma ("5.4"), we may assume that G is $\Delta(G)$ -regular ($\forall v \in V(G), \deg_G(v) = \Delta(G)$).

- (i) $\exists v \in V(G)$ such that $G - v$ is disconnected. Then $G - v$ has at least two components. Let $C_1, \dots, C_k$ be the components of $G - v$, and let $G_i = G[V(C_i) \cup \{v\}]$. G_i is connected, $\Delta(G_i) = \Delta(G)$ and since $\deg_{G_i}(v) < \deg_G(v) = \Delta(G) \implies$ (by Lemma "5.4") we can color G_i with $\Delta(G)$ colors. Fix such a coloring and then permute the colors s.t. v always has the same color. Then this is a coloring of G with $\Delta(G)$ colors.

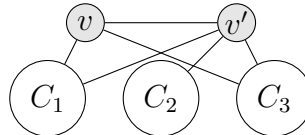


- (ii) Wlog G is connected.

Assume not (1) and $\exists v, v' \in V(G)$ s.t. $G - v - v'$ is disconnected.

Let $C'_1, \dots, C'_k$ be the components of $G - v - v'$. Let $G_i = G[V(C'_i) \cup \{v, v'\}]$. Let G'_i be the graph obtained from G_i by adding $\{v, v'\}$ if it is not already there.

Observe that $\forall i \in \{1, \dots, k\}$, v has a neighbour in C_i (say v_i).



$$\deg_{G'_i}(v) = \begin{cases} \deg_G(v) - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \in E(G) \\ \deg_G(v) + 1 - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \notin E(G) \end{cases} \leq \begin{cases} \Delta(G) - (k-1) & \{v, v'\} \in E(G) \\ \Delta(G) - (k-2) & \{v, v'\} \notin E(G) \end{cases}$$

Same goes for $\deg_{G'_i}(v')$

If $\deg_{G'_i}(v) < \Delta(G)$ or $\deg_{G'_i}(v') < \Delta(G)$, then we can $\Delta(G)$ -color G_i by Lemma (5.4).

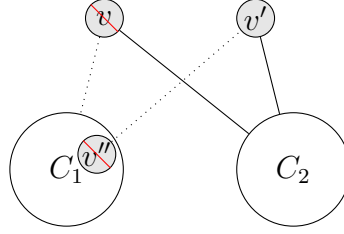
If $G'_1, \dots, G'_k$ have a $\Delta(G)$ -coloring $\implies$ can combine them to a $\Delta(G)$ -coloring

of G .

Suppose $\exists i \in \{1, \dots, k\}$ s.t. $\deg_{G'_i}(v) = \deg_{G'_i}(v') = \Delta(G)$. Wlog $i = 2$. Then $k = 2, \{v, v'\} \notin E(G)$ ($\deg_{G'_i}(v) < \Delta(G)$ if $k \geq 3$) and $|N_{C_1}(v)| = |N_{C_2}(v')| = 1$.

Let $v'' \in V(C_1)$ be the neighbour of v

Observation: $\bar{G} = G - v - v''$ is not connected. $\bar{G}$ has components $D_1 := C_1 - v''$ and $D_2 = [V(C_2) \cup \{v'\}]$. v has $\Delta(G) - 1$ neighbours in D_2 and v'' has $\Delta(G) - 1$ neighbours in D_1 . Use the above coloring argument on $G - v - v''$.



(iii) Wlog G is connected.

Assume not (1) and not (2). Then $\forall v, v' \in V(G)$, $G - v - v'$ is connected.

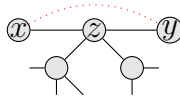
G not complete $\implies \exists x, x' \in V(G), x \neq x'$, non adjacent. Let P be a shortest path from x to x' . Let z be the neighbour of x on P and y be the neighbour of z on P distinct from x . As P is a shortest path, x and y are nonadjacent.

Fix a spanning tree of $G - x - y$ rooted at z . Set $v_1 = x, v_2 = y$ and $v_3, \dots, v_n$ is a listing of the remaining vertices of G s.t. every vertex in the tree precedes its parents. In particular $v_n = z$. Now we give the same color to v_1 and v_2 .

Suppose $v_1, v_2, \dots, v_{i-1}$ are colored already.

If $i < n \implies v_i$ is adjacent to at least one vertex in $\{v_{i+1}, \dots, v_n\}$ namely its parent. So v_i has at most $\Delta(G) - 1$ neighbours that already have a color, so we have 1 color left.

For $i = n, z$ has two neighbours with same color (x, y) , so again there is a color not present among the neighbours of z , which we assign to z .



□

Theorem 5.6 (4-Color Theorem). *Every planar graph is 4-colorable.*

Theorem 5.7 (5-Color Theorem). *Every planar graph is 5-colorable.*

Proof. Induction on $n = |V(G)|$. Fix a plane drawing of G .

$n = 1, 2$: trivial.

Let $n \geq 3$:

Suppose $\exists v \in V(G)$ $\deg_G(v) \leq 4$. Then color $G - v$ by ind. with 5 colors. The neighbours of v use at most 4 colors, so there is a color we can assign to v .

Now assume that G is maximal.

From the proof of Lemma ("3.4"), it follows that G is a triangulation. By Ex 3.2 $\exists w \in V(G)$ s.t. $\deg_G(w) = 5$. Let $v_1, \dots, v_5$ be the neighbours of w , listed in cyclic ordering around w . Then $\{v_1, v_3\}$ or $\{v_2, v_4\}$ is not an edge in $E(G)$, Wlog $\{v_1, v_3\}$.

Let G' be the plane graph obtained by removing w, v_1, v_3 and inserting w' at the position of w . Join every neighbour u of v_1 in G to w' by inserting an edge along $\{u, v_1\}$ and then along $\{v_1, w\}$ (same for the neighbours of v_3 but put only one neighbour of both v_1, v_3). $|V(G')| = n - 2$. G' is planar. By induction, there exists a 5-coloring $c' : V(G') \rightarrow \{1, \dots, 5\}$ of G' .

$$\text{Let } c : V(G) \rightarrow \{1, \dots, 5\} \text{ with } c(v) = \begin{cases} c'(v), & v \in V(G) \setminus \{w, v_1, v_3\} \\ c(v_1) = c'(w') \\ c(v_3) = c'(w') \\ c(w) \text{ is any color } \notin \{c(v_i) : i = 1, \dots, 5\} \end{cases}$$

Then c is a 5-coloring of G . □

Definition 5.8. The **clique number** of a graph G , denoted $\omega(G)$, is the order of the largest complete subgraph of G .

Remark 5.9. Clearly, $\omega(G) \leq \lambda(G) \leq \Delta(G) + 1$.

Theorem 5.10. Let $k \geq 2$. There exists a graph G with $\omega(G) = 2$ and $\chi(G) > k$.

Definition 5.11. The **girth** of a graph G is the length of the shortest cycle in G .

Theorem 5.12. For every $k \geq 1$, there exists a graph G with $g(G) > k$ and $\chi(G) > k$.

Proof. postponed □

Remark 5.13. 5.12 $\implies$ 5.10 as $g(G) > 3 \implies G$ has no triangles $\implies \omega(G) \leq 2$.

Corollary 5.14. $\chi(G) \leq \lceil \frac{\omega(G) + \Delta(G) + 1}{2} \rceil$

5.1 Perfect Graphs

Definition 5.15. A graph G is **perfect** if for every induced subgraph G' of G , we have $\chi(G') = \omega(G')$.

Example 5.16. complete graphs and bipartite graphs are perfect.

Theorem 5.17 (weak perfect Graph). *A graph G is perfect if and only if $\overline{G}$ is perfect.*

($\overline{G}$ is the complement of G which is the graph with the same vertex set as G and two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G)

Theorem 5.18 (strong perfect Graph). *A graph G is perfect if and only if G or its complement does not contain an odd cycle of length at least 5 as an induced subgraph.*

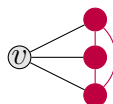
Definition 5.19. A graph is chordal if it does not contain an induced cycle of length at least 4.

Remark 5.20. Induced subgraphs of chordal graphs are chordal.

Theorem 5.21. *Every chordal graph is perfect.*

Proof. Use Theorem 5.17. □

Definition 5.22. A vertex v in a graph G is **simplicial** if $N_G(v)$ induces a complete graph.



Lemma 5.23. *Let G be a graph. If G is chordal, then it has a simplicial vertex.*

Proof. Induction on $n := |V(G)|$. Statement is true for $n = 1$ and if G is complete. Now $n \geq 2$ and G not complete.

Choose $U \subseteq V(G)$ of maximal size s.t. $G[U]$ (= subgraph induced by the vertices in U) is connected and $U \cup N_G(U) \neq V(G)$ ($N_G(U) = \bigcup_{u \in U} N_G(u) \setminus U$).

If $u, u' \in V(G)$ are non-adjacent, $u \neq u'$, then $U = \{u\}$ satisfies the above condition.

Let $W = V(G) \setminus (U \cup N_G(U))$, then $W \neq \emptyset$. Every $x \in N_G(U)$ is adjacent to all vertices in W , otherwise we could add x to U $\nmid$ maximality of U . $G[W]$ is chordal and by induction W has a simplicial vertex v .

Claim: v is simplicial in G .

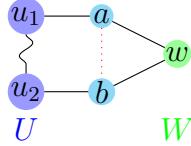
We show that $N_G(U) := N(U)$ is a clique.

Suppose there are $a, b \in N(U)$, a and b non-adjacent, $a \neq b$. Let $w \in W \implies w$ is adjacent to both a and b . Let P be a shortest path from a to b with inner vertices in U (exists since $G[U]$ is connected and both a, b have neighbours in U).

Then concatenating P with $(b, \{b, w\}, w, \{a, w\}, a)$ yields a cycle in G that does not have a chord, the cycle has length $\geq 4 \nmid G$ chordal.

$\implies N(U)$ is a clique.

$\implies N_G(v) = N(U)$ is a clique $\implies v$ is simplicial in G .



□

Theorem 5.24. *Let G be a graph. Then G is chordal iff G has a perfect elimination ordering, i.e. an ordering $v_1, \dots, v_n$ of $V(G)$ s.t. $\forall i \in \{1, \dots, n\}$, the neighbours of v_i in $\{v_{i+1}, \dots, v_n\}$ induce a complete graph.*

Proof. " $\implies$ " Let v_1 be a simplicial vertex in G (exists by Lemma 5.23). Proceed inductively on $G - v_1$ (which is chordal).

" $\impliedby$ " Let C be an induced cycle of G of length ≥ 3 . Let v_i be a PEO. Let i be minimal s.t. $v_i \in V(C)$. Then the two neighbours of v_i on C are v_{i+j} and v_{i+l} for some $j, l > 0$. Since the neighbours of v_i in $\{v_{i+1}, \dots, v_n\}$ induce a complete graph, v_{i+j} and v_{i+l} are adjacent, so $|V(C)| = 3$. Thus G is chordal. □

5.2 EdgeColoring

Definition 5.25. Let $k \in \mathbb{N}$. A (proper) **k -edge-coloring** of a graph G is a function $c : E(G) \rightarrow \{1, 2, \dots, k\}$ such that no two adjacent edges have the same color.

The **chromatic index (edge-chromatic-number)** $\chi'(G)$ of a graph G is the smallest k such that G has a proper k -edge-coloring.

Remark 5.26. $\Delta(G) \leq \chi'(G)$

Theorem 5.27 (Vizing's theorem). *Every graph G has a proper edge coloring with at most $\Delta(G) + 1$ colors. Thus $\chi'(G) \in \{\Delta(G), \Delta(G) + 1\}$.*

Proof. not here. □

6 Ramsey Theory

In this chapter, if we talk about coloring, we always mean a non-proper coloring (think of a child painting the graph randomly).

Example 6.1.

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